

Weather Effects on Marathon Performance: A Big Data Analysis

Zijian Wang

Xiaohan Zhang

Yuanyuan Zhang

Abstract

Marathon running performance is strongly influenced by environmental conditions, particularly weather. Prior research has shown that race-day temperature plays a key role in shaping finishing times, often exhibiting non-linear effects (El Helou et al. (2012); Vihma (2009); Mantzios et al. (2021)). This project proposes a large-scale big data analysis of the relationship between weather and marathon performance using a publicly available marathon dataset. Using distributed data processing technologies, we will analyze runner-level marathon results together with the dataset’s race-day weather summary component to explore how temperature and related meteorological variables affect performance across different runner groups. The project will leverage Apache Spark and related components of the Hadoop ecosystem to perform scalable data ingestion, processing, and analysis. The expected outcome is a reproducible big data pipeline and interpretable results that quantify the impact of weather on marathon outcomes.

Problem

Weather conditions vary substantially across marathon events and years, and these variations can meaningfully affect runners’ performance and safety. While existing studies confirm that higher temperatures are generally associated with slower finishing times, many analyses rely on relatively small datasets or single-machine statistical workflows. Large-scale empirical analyses have documented a U-shaped relationship between temperature and performance, with optimal results typically observed in cool conditions (El Helou et al. (2012); Vihma (2009)). The availability of a large, longitudinal marathon dataset that combines runner-level performance data with race-day weather summaries provides an opportunity to revisit this problem using big data technologies. The key problem addressed in this project is how to efficiently integrate and analyze a large runner-level performance dataset together with its corresponding race-day weather summaries to better understand the magnitude and structure of weather effects—especially temperature—on marathon performance.

Intended Contribution

The intended contribution of this project is twofold. First, we aim to extend earlier marathon-weather analyses by adopting a true big data approach, using distributed storage and computation tools instead of conventional single-machine workflows such as those implemented in R. Second, we

seek to provide clear, data-driven estimates of how weather variables—particularly temperature—affect marathon finishing times across different performance groups (e.g., faster vs. slower runners). By the end of the project, we expect to deliver a scalable analytics pipeline, well-documented code, and a set of empirical findings that can be compared with results from prior sport science literature.

Methodology

Data Sets

We will use the Berlin Marathons Data (1974–2019) available on Kaggle (David (2023)) as the sole data source for this project. This dataset contains two related components: (1) runner-level results data with approximately 885,000 observations, including variables such as year, gender, age, and official finishing time; and (2) annual race-day weather summaries including average, minimum, and maximum temperature, precipitation, sunshine hours, cloud cover, and atmospheric pressure. The two components can be linked by race year, allowing each runner’s finishing time to be associated with the corresponding race-day weather conditions. The size and longitudinal structure of this dataset make it well-suited for distributed processing and large-scale analytical workflows.

Tools and Technologies

The project will be implemented using a modern big data technology stack:

- **Programming languages:** Python (primary), with optional use of Scala for Spark
- **Big data frameworks:** Apache Hadoop (HDFS), Apache Spark
- **Data processing:** Spark-based large-scale joins and transformations
- **Optional:** Kafka for streaming ingestion if time and learning permit

Computational Resources

All computations will be performed using local or cloud-based resources capable of running Hadoop and Spark. The datasets are sufficiently large to justify distributed processing but manageable within the available resources.

Analysis Plan

1. **Data ingestion and cleaning:** Load marathon and weather datasets into HDFS-compatible storage; handle missing values and ensure consistent formats.
2. **Data integration:** Join runner-level results with weather summaries using the race year as the key identifier.
3. **Exploratory data analysis (EDA):** Use Spark-based aggregations and visual summaries to explore distributions of finishing times and weather conditions.
4. **Modeling and analysis:** Quantify the relationship between temperature and finishing time using scalable regression or grouped summary analyses within Spark.
5. **Subgroup analysis:** Compare weather effects across performance strata (e.g., elite vs. recreational runners).

Expected Results

We expect to observe a clear association between race-day temperature and marathon finishing times, consistent with prior findings: cooler temperatures are likely associated with faster performances, while warmer conditions lead to slower times. This expectation is consistent with prior multi-race analyses that report measurable performance declines for each 1°C increase above optimal conditions (Mantzios et al. (2021); Weiss et al. (2022)). We also anticipate that the magnitude of the effect will differ across runner groups. From a technical perspective, we expect to produce a working big data pipeline that demonstrates effective use of Hadoop and Spark for large-scale sports analytics.

Team Responsibilities

This project will be completed as a team of three students. Responsibilities are divided as follows:

Team	
Member	Responsibilities
Zijian Wang	Coordination, background research, problem formulation, coding support, and problem/contribution writing
Xiaohan Zhang	Data ingestion, Spark-based processing, main analysis pipeline, and report writing
Yuanyuan Zhang	Exploratory data analysis, visualization, interpretation of findings, and contribution to the final report

Key References

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- Mantzios, K., Ioannou, L. G., Panagiotaki, Z., Ziaka, S., Périard, J. D., Racinais, S., Nybo, L., & Flouris, A. D. (2021). Effects of weather parameters on endurance running performance: Discipline specific analysis of 1258 races. *Medicine & Science in Sports & Exercise*, 54. <https://doi.org/10.1249/MSS.0000000000002769>
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